INDIVIDUAL WEEKLY PROJECT REPORTS

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Role: Exploratory Data Analysis and Data Visualization
Project Title: Flight Delay Analysis and Visualization

WEEK 7 – Visual Interpretation of Correlation and PCA

During the seventh week, I focused on understanding the statistical analysis results and exploring how they could be represented visually. The team examined the relationship between numerical variables using correlation analysis and also explored PCA.

One important result discussed during this stage was the strong positive correlation of approximately 0.971 between departure delay and arrival delay. I understood that this relationship indicated that flights with greater departure delays generally tended to experience greater arrival delays.

I explored how a scatter plot could be used to communicate this relationship visually. A scatter plot is useful because it displays

individual observations and helps viewers understand the direction and strength of a relationship.

I also familiarized myself with PCA and its purpose in reducing the dimensionality of numerical data. I understood that PCA transforms the original variables into principal components that capture important variation in the dataset.

I coordinated with the analysis team to understand the results and discussed how the findings could be included in the project presentation using suitable visual representations.

Outcome and Learning:

This week improved my understanding of statistical visualization and interpretation. I learned how visual representations can help explain relationships between variables and make statistical results easier to communicate.